ALLEN CAREER INSTITUTE
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KOTA (RAJASTHAN)



INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

d- Block Compounds

English Medium

EXERCISE # O-1

SELECT ONLY ONE IS CORRECT OPTIONS :

General Properties of d-block

1. $N_2(g) + 3H_2(g) \underset{}{\overset{Fe + Mo}{\rightleftharpoons}} 2NH_3(g)$; Haber's process, Mo is used as

(A) a catalyst
(B) a catalytic promoter
(C) an oxidising agent
(D) as a catalytic poison

DB0001

2. An ornamental of gold having 75% of gold, it is of ..... carat.

(A) 18 (B) 16 (C) 24 (D) 20

DB0002

3. Transition elements having more tendency to form complex than representative elements (s and p-block elements) due to -

(A) availability of d-orbitals for bonding
(B) variable oxidation states are not shown by transition elements
(C) all electrons are paired in d-orbitals
(D) *f*-orbitals are available for bonding

DB0003

4. A compound of mercury used in cosmetics, in Ayurvedic and Yunani medicines and known as Vermilion is -

(A) $HgCl_2$ (B) HgS (C) Hg_2Cl_2 (D) HgI

DB0004

5. Transition elements are usually characterised by variable oxidation states but Zn does not show this property because of

(A) completion of np-orbitals
(B) completion of (n–1)d orbitals
(C) completion of ns-orbitals
(D) inert pair effect

DB0005

6. The d-block element which is a liquid at room temperature, having high specific heat, less reactivity than hydrogen and its chloride (MX_2) is volatile on heating is

(A) Cu (B) Hg (C) Ce (D) Pm

DB0006

7. Coinage metals show the properties of

(A) typical elements
(B) normal elements
(C) inner-transition elements
(D) transition element

DB0007

8. The transition metal used in X-rays tube is

(A) Mo (B) Ta (C) Tc (D) Pm

DB0008

9. The higher oxidation states of transition elements are found to be in the combination with A and B, which are

(A) F, O (B) O, N (C) O, Cl (D) F, Cl

DB0009

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10. The metals present in insulin and haemoglobin are respectively

(A) Zn, Hg (B) Zn, Fe (C) Co, Fe (D) Mg, Fe

DB0010

11. A metal M which is not affected by strong acids like conc. HNO_3, conc. H_2SO_4 and conc. solution of alkalies like NaOH, KOH forms MCl_3 which finds use for toning in photography. The metal M is

(A) Ag (B) Hg (C) Au (D) Cu

DB0011

12. Manganese steel is used for making railway tracks because

(A) it is hard with high percentage of Mn

(B) it is soft with high percentage of Mn

(C) it is hard with small concentration of manganese with impurities

(D) it is soft with small concentration of manganese with impurities

DB0012

13. Transition elements in lower oxidation states act as Lewis acid because

(A) they form complexes

(B) they are oxidising agents

(C) they donate electrons

(D) they do not show catalytic properties

DB0013

14. The electrons which take part in order to exhibit variable oxidation states by transition metals are

(A) ns only

(B) (n–1)d only

(C) ns and (n–1)d only but not np

(D) (n–1)d and np only but not ns

DB0014

15. Solution of MnO_4^- is purple-coloured due to

(A) d-d-transition

(B) charge transfer from O to Mn

(C) due to both d-d-transition and charge transfer

(D) none of these

DB0015

16. An element of 3d-transition series shows two oxidation states x and y, differ by two units then

(A) compounds in oxidation state x are ionic if x > y

(B) compounds in oxidation state x are ionic if x < y

(C) compounds in oxidation state y are covalent if x < y

(D) compounds in oxidation state y are covalent if y < x

DB0016

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Compounds of d-block

17. (T) $\xrightarrow{\text{compd (U)} + \text{conc.}H_2SO_4}$ $(V)_{\text{Red gas}}$ $\xrightarrow{NaOH+AgNO_3}$ $(W)_{\text{Red ppt.}}$ $\xrightarrow{NH_3 \text{ soln.}}$ (X)

imparts violet colour in the flame test

$(W)_{\text{Red ppt.}} \xrightarrow{\text{dil. HCl}} (Y)_{\text{white ppt.}}$

(U) $\xrightarrow[\Delta]{NaOH}$ $(Z)_{gas}$ (gives white fumes with HCl)

sublimes on heating

Identify (T) to (Z).

(A) T = $KMnO_4$, U = HCl, V = Cl_2, W = HgI_2, X = $Hg(NH_2)NO_3$, Y = Hg_2Cl_2, Z = N_2

(B) T=$K_2Cr_2O_7$, U=NH_4Cl, V=CrO_2Cl_2, W=Ag_2CrO_4, X=$[Ag(NH_3)_2]^+$, Y=AgCl,Z = NH_3

(C) T = K_2CrO_4, U = KCl, V = CrO_2Cl_2, W = HgI_2, X = Na_2CrO_4, Y = $BaCO_3$, Z = NH_4Cl

(D) T = K_2MnO_4, U = NaCl, V = CrO_3, W = $AgNO_2$, X = $(NH_4)_2CrO_4$, Y = $CaCO_3$, Z = SO_2

DB0017

18. The number of moles of acidified $KMnO_4$ required to convert one mole of sulphite ion into sulphate ion is

(A) 2/5 (B) 3/5 (C) 4/5 (D) 1

DB0018

19. $Cr_2O_7^{2-} \underset{Y}{\overset{X}{\rightleftharpoons}} 2CrO_4^{2-}$, X and Y are respectively

(A) X = OH^-, Y = H^+ (B) X = H^+, Y = OH^-

(C) X = OH^-, Y = H_2O_2 (D) X = H_2O_2, Y = OH^-

DB0019

20. CrO_3 dissolves in aqueous NaOH to give

(A) $Cr_2O_7^{2-}$ (B) CrO_4^{2-} (C) $Cr(OH)_3$ (D) $Cr(OH)_2$

DB0020

21. During estimation of oxalic acid Vs $KMnO_4$, self indicator is

(A) $KMnO_4$ (B) oxalic acid (C) K_2SO_4 (D) $MnSO_4$

DB0021

22. Acidified chromic acid + H_2O_2 $\xrightarrow{\text{Org.solvent}}$ X + Y , X and Y are (blue colour)

(A) CrO_5 and H_2O (B) Cr_2O_3 and H_2O (C) CrO_2 and H_2O (D) CrO and H_2O

DB0022

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23. $\underset{\text{(diatomic covalent molecule)}}{Y} \xleftarrow{KI} CuSO_4 \xrightarrow{\text{dil } H_2SO_4} X(\text{Blue colour})$, X and Y are

(A) $X = I_2$, $Y = [Cu(H_2O)_4]^{2+}$ (B) $X = [Cu(H_2O)_4]^{2+}$, $Y = I_2$

(C) $X = [Cu(H_2O)_4]^{+}$, $Y = I_2$ (D) $X = [Cu(H_2O)_5]^{2+}$, $Y = I_2$

DB0023

24. $(NH_4)_2Cr_2O_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is

(A) Cr_2O_3 (B) CrO_2 (C) Cr_2O_4 (D) CrO_3

DB0024

25. Iron becomes passive by ..... due to formation of .....

(A) dil. HCl, Fe_2O_3 (B) 80% conc. HNO_3, Fe_3O_4

(C) conc. H_2SO_4, Fe_3O_4 (D) conc. HCl, Fe_3O_4

DB0025

26. Bayer's reagent used to detect olifinic double bond is

(A) acidified $KMnO_4$ (B) aqueous $KMnO_4$

(C) 1% alkaline $KMnO_4$ solution (D) $KMnO_4$ in benzene

DB0026

27. $MnO_4^- + xe^- \xrightarrow{\text{(Alkaline medium)}} MnO_4^{2-}$

$\xrightarrow{+ye^- \text{ (Acidic medium)}} Mn^{2+}$

$\xrightarrow{+ze^- \text{ (Neutral medium)}} MnO_2$

x, y and z are respectively

(A) 1, 2, 3 (B) 1, 5, 3 (C) 1, 3, 5 (D) 5, 3, 1

DB0027

28. $Cu + \text{conc. } HNO_3 \xrightarrow[\text{(hot)}]{} Cu(NO_3)_2 + X$ (oxide of nitrogen); then X is

(A) N_2O (B) NO_2 (C) NO (D) N_2O_3

DB0028

29. $CuSO_4$ solution reacts with excess KCN to give

(A) $Cu(CN)_2$ (B) CuCN (C) $K_2[Cu(CN)_2]$ (D) $K_3[Cu(CN)_4]$

DB0029

30. Pick out the incorrect statement:

(A) MnO_2 dissolves in conc. HCl, but does not form Mn^{4+} ions

(B) MnO_2 react with hot concentrated H_2SO_4 liberating oxygen

(C) K_2MnO_4 is formed when MnO_2 in fused KOH is oxidised by air, KNO_3 , PbO_2 or $NaBiO_3$

(D) Decomposition of acidic $KMnO_4$ is not catalysed by sunlight.

DB0030

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31. 1 mole of Fe^{2+} ions are oxidised to Fe^{3+} ions with the help of (in acidic medium)
(A) 1/5 moles of $KMnO_4$ (B) 5/3 moles of $KMnO_4$
(C) 2/5 moles of $KMnO_4$ (D) 5/2 moles of $KMnO_4$

DB0031

32. To an acidified dichromate solution, a pinch of Na_2O_2 is added and shaken. What is observed:
(A) blue colour (B) Orange colour changing to green
(C) Copious evolution of oxygen (D) Bluish - green precipitate

DB0032

33. The rusting of iron is formulated as $Fe_2O_3 \cdot xH_2O$ which involves the formation of
(A) Fe_2O_3 (B) $Fe(OH)_3$ (C) $Fe(OH)_2$ (D) $Fe_2O_3 + Fe(OH)_3$

DB0033

34. Solid $CuSO_4 \cdot 5H_2O$ having covalent, ionic as well as co-ordinate bonds. Copper atom/ion forms ..... co-ordinate bonds with water.
(A) 1 (B) 2 (C) 3 (D) 4

DB0034

35. $KMnO_4 + HCl \longrightarrow H_2O + X(g)$, X is a
(acidified)
(A) red liquid (B) violet gas
(C) greenish yellow gas (D) yellow-brown gas

DB0035

36. Purple of cassius is:
(A) Pure gold (B) Colliodal solution of gold
(C) Gold (I) hydroxide (D) Gold (III) chloride

DB0036

37. Amongst the following species, maximum covalent character is exhibited by
(A) $FeCl_2$ (B) $ZnCl_2$ (C) $HgCl_2$ (D) $CdCl_2$

DB0037

38. Number of moles of $SnCl_2$ required for the reduction of 1 mole of $K_2Cr_2O_7$ into Cr_2O_3 is (in acidic medium)
(A) 3 (B) 2 (C) 1 (D) 1/3

DB0038

39. Pick out the incorrect statement:

(A) MnO_4^{2-} is quite strongly oxidizing and stable only in very strong alkalies. In dilute alkali, neutral solutions, it disproportionates.

(B) In acidic solutions, MnO_4^- is reduced to Mn^{2+} and thus, $KMnO_4$ is widely used as oxidising agent

(C) $KMnO_4$ does not acts as oxidising agent in alkaline medium

(D) $KMnO_4$ is manufactured by the fusion of pyrolusite ore with KOH in presence of air or KNO_3, followed by electrolytic oxidation in strongly alkaline solution.

DB0039

40. The aqueous solution of $CuCrO_4$ is green because it contains
(A) green Cu^{2+} ions (B) green CrO_4^{2-} ions
(C) blue Cu^{2+} ions and green CrO_4^{2-} ions (D) blue Cu^{2+} ions and yellow CrO_4^{2-} ions

DB0040

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41. In nitroprusside ion, the iron exists as Fe^{2+} and NO as NO^+ rather than Fe^{3+} and NO respectively. These forms of ions are established with the help of

(A) magnetic moment in solid state
(B) thermal decomposition method
(C) by reaction with KCN
(D) by action with K_2SO_4

DB0041

42. Which of the following reaction is possible at anode?

(A) $2Cr^{3+} + 7H_2O \longrightarrow Cr_2O_7^{2-} + 14\ H^+$
(B) $F_2 \longrightarrow 2\ F^-$
(C) $\frac{1}{2}O_2 + 2H^+ \longrightarrow H_2O$
(D) None of these

DB0042

43. Colourless solutions of the following four salts are placed separately in four different test tubes and a strip of copper is dipped in each one of these. Which solution will turn blue?

(A) KNO_3 (B) $AgNO_3$ (C) $Zn(NO_3)_2$ (D) $ZnSO_4$

DB0043

44. When acidified $KMnO_4$ is added to hot oxalic acid solution, the decolourization is slow in the beginning, but becomes very rapid after some time. This is because:

(A) Mn^{2+} acts as autocatalyst
(B) CO_2 is formed as the product
(C) Reaction is exothermic
(D) MnO_4^- catalyses the reaction

DB0044

45. Metre scales are made-up of alloy

(A) invar (B) stainless steel (C) elektron (D) magnalium

DB0045

46. The Ziegler-Natta catalyst used for polymerisation of ethene and styrene is $TiCl_4 + (C_2H_5)_3Al$, the catalysing species (active species) involved in the polymerisation is

(A) $TiCl_4$ (B) $TiCl_3$ (C) $TiCl_2$ (D) TiCl

DB0046

47. 'Bordeaux mixture' is used as a fungicide. It is a mixture of

(A) $CaSO_4 + Cu(OH)_2$
(B) $CuSO_4 + Ca(OH)_2$
(C) $CuSO_4 + CaO$
(D) $CuO + CaO$

DB0047

48. Peacock ore is:

(A) FeS_2 (B) $CuFeS_2$ (C) $CuCO_3.Cu(OH)_2$ (D) Cu_5FeS_4

DB0048

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EXERCISE # O-2

SELECT MORE THAN ONE IS CORRECT OPTIONS

General Properties of d-block

1. Potash alum is a double salt, its aqueous solution shows the characteristics of

(A) Al^{3+} ions (B) K^+ ions (C) SO_4^{2-} ions (D) Al^{3+} ions but not K^+ ions

DB0049

2. Addition of non-metals like B and C to the interstitial sites of a transition metal results the metal

(A) of more ductability (B) of less ductability

(C) less malleable (D) of more hardness

DB0050

3. Mercury is a liquid at 0°C because of

(A) very high ionisation energy (B) weak metallic bonds

(C) high heat of hydration (D) high heat of sublimation

DB0051

4. The correct statement(s) about transition elements is/are

(A) the most stable oxidation state is +3 and its stability decreases across the period

(B) transition elements of 3d-series have almost same atomic sizes from Cr to Cu

(C) the stability of +2 oxidation state increases across the period

(D) some transition elements like Ni, Fe, Cr may show zero oxidation state in some of their compounds

DB0052

5. The ionisation energies of transition elements are

(A) less than p-block elements (B) more than s-block elements

(C) less than s-block elements (D) more than p-block elements

DB0053

6. The metal(s) which does/do not form amalgam is/are

(A) Fe (B) Pt (C) Zn (D) Ag

DB0054

7. Which of the following statements concern with d-block metals?

(A) compounds containing ions of transition elements are usually coloured

(B) Zinc has lowest melting point among 3d-series elements

(C) they show variable oxidation states, which differ by two units only

(D) they easily form complexes

DB0055

8. The highest oxidation state among transition elements is

(A) + 7 by Mn (B) + 8 by Os (C) + 8 by Ru (D) + 7 by Fe

DB0056

9. Amphoteric oxide(s) is/are

(A) Al_2O_3 (B) SnO (C) ZnO (D) Fe_2O_3

DB0057

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10. The catalytic activity of transition elements is related to their
(A) variable oxidation states (B) free suface valencies
(C) complex formation ability (D) magnetic moment

DB0058

11. In the equation: $M + 8CN^- + 2H_2O + O_2 \longrightarrow 4[M(CN)_2]^- + 4OH^-$, metal M is
(A) Ag (B) Au (C) Cu (D) Hg

DB0059

12. $CuSO_4(aq) + 4NH_3 \longrightarrow X$, then X is
(A) $[Cu(NH_3)_4]^{2+}$ (B) paramagnetic
(C) coloured (D) of a magnetic moment of 1.73 BM

DB0060

13. Amphoteric oxide(s) of Mn is/are
(A) MnO_2 (B) Mn_3O_4 (C) Mn_2O_7 (D) MnO

DB0061

14. The lanthanide contraction is responsible for the fact that
(A) Zr and H*f* have same atomic sizes (B) Zr and H*f* have same properties
(C) Zr and H*f* have different atomic sizes (D) Zr and H*f* have different properties

DB0062

15. Ion(s) having non zero magnetic moment (spin only) is/are
(A) Sc^{3+} (B) Ti^{3+} (C) Cu^{2+} (D) Zn^{2+}

DB0063

Compounds of d-block

16. Correct statement(s) is/are
(A) an acidified solution of $K_2Cr_2O_7$ liberates iodine from KI
(B) $K_2Cr_2O_7$ is used as a standard solution for estimation of Fe^{2+} ions
(C) in acidic medium, M = N/6 for $K_2Cr_2O_7$
(D) $(NH_4)_2Cr_2O_7$ on heating decomposes to yield Cr_2O_3 through an endothermic reaction

DB0064

17. Interstitial compounds are formed by
(A) Co (B) Ni (C) Fe (D) Ca

DB0065

18. Acidified $KMnO_4$ can be decolourised by
(A) SO_2 (B) H_2O_2 (C) $FeSO_4$ (D) $Fe_2(SO_4)_3$

DB0066

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EXERCISE # S-1

NUMERICAL GRID TYPE QUESTIONS :

1. When mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc.H_2SO_4 then compound X is formed. What is the oxidation state of central atom of X.

DB0067

2. Number of ions which gives blue colour in aqueous state :

V^{+4}, Ni^{+2}, Ti^{+3}, Co^{+2}, Fe^{+3}, Cu^{+2}

DB0068

3. Define the oxidation states of Mn in product of the given reaction

$3K_2MnO_4+2H_2O+4CO_2 \rightarrow 2X+Y+4KHCO_3$

If the oxidation state of Mn in product X and Y are n_1 and n_2 respectively. Then find out the value of (n_1+n_2) -

DB0069

4. Find number of metal ion which can produce high spin and low spin octahedral complex :

Sc^{+3}, Ti^{+3}, V^{+3}, Cr^{+3}, Mn^{+3}, Fe^{+3}, Co^{+3}, Ni^{+2}

DB0070

5. How many non-axial d-orbitals are involved in hybridisation of CrO_2Cl_2.

DB0071

6. Find the number of species from the following which has magnetic moment value of 1.73 B.M.

Fe^{2+}, Cu^{2+}, Ni^{2+}, NO_2, NO_2^-, Ti^{3+}

DB0072

7. Total no. of moles of Mohr's salt are required for per mole of dichromate ions during volumetric analysis.

DB0073

8. Find number of reaction(s) in which no redox change takes place -

(I) $BaCl_2 + Na_2CrO_4 \xrightarrow{CH_3COOH}$

(II) $K_2Cr_2O_7 + NaOH \longrightarrow$

(III) $Cr_2O_7^{2-} + NO_3^- \xrightarrow{H^+}$

(IV) $Cr_2O_7^{2-} + C_2H_5OH \xrightarrow{H^+}$

DB0074

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EXERCISE # S-2

COMPREHENSTION TYPE QUESTIONS :

Comprehension # 01 to 04

Transition metal and their compounds are used as catalysts in industry and in biological system. For example, in the Contact process, vanadium compounds in the +5 state (V_2O_5 or VO_3^-) are used to oxidise SO_2 to SO_3:

$$SO_2 + \frac{1}{2}O_2 \xrightarrow{V_2O_5} SO_3$$

It is thought that the actual oxidation process takes place in two stages. In the first step, V^{5+} in the presence of oxide ions converts SO_2 to SO_3. At the same time, V^{5+} is reduced to V^{4+}.

$$2V^{5+} + O^{2-} + SO_2 \longrightarrow 2V^{4+} + SO_3$$

In the second step, V^{5+} is regenerated from V^{4+} by oxygen :

$$2V^{4+} + \frac{1}{2}O_2 \longrightarrow 2V^{5+} + O^{2-}$$

The overall process is, of course, the sum of these two steps:

$$SO_2 + \frac{1}{2}O_2 \longrightarrow SO_3$$

1. Transition metals and their compounds catalyse reactions because **:**

(A) They have competely filled s-subshell

(B) They have a comparable size due to poor shielding of d-subshell

(C) They introduce an entirely new reaction mechanism with a lower activation energy

(D) They have variable oxidation states differ by two units

DB0075

2. During the course of the reaction **:**

(A) Catalyst undergoes changes in oxidation state

(B) Catalyst increases the rate constant

(C) Catalyst is regenerated in its orginal form when the reactants form the products

(D) All are correct.

DB0075

3. Catalytic activity of transition metals depends on **:**

(A) Their ability to exist in different oxidation states

(B) The size of the metal atoms

(C) The number of empty atomic orbitals available

(D) None of these

DB0075

4. Which of the following ion involved in the above process will show paramagnetism?

(A) V^{5+} (B) V^{4+} (C) O^{2-} (D) VO_3^-

DB0075

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Comprehension # 05 & 06

(X) is very important laboratory reagent which is prepared by its naturally occuring ore which is called pyrolusite. Pyrolusite when fused with alkali in the presence of O_2, green compound **(Y)** is produced. **(Y)** is converted into **(X)** by electrolysis or by using ozone.

5. On small scale **(X)** is prepared by disproportion of **(Y)** in acidic solution. Which of the following is produced by disproportion of **(Y)** in slight alkaline solution.

(A) $KMnO_4$, Mn^{+2} (B) $KMnO_4$, MnO_2

(C) MnO_2, Mn^{+2} (D) K_2MnO_4, Mn^{+2}

DB0076

6. Select the correct statements :

(A) **(X)** is tetrahedral & diamagnetic

(B) **(Y)** is tetrahedral & paramagnetic

(C) **(X)** produce dimangnese hepta oxide (oily liquid) with conc. H_2SO_4

(D) All are correct

DB0076

Comprehension # 07 to 09

Due to availability of vacant orbitals of sufficiently low energy, d-block elements form complexes, d-block elements have different properties such as- catalytic, magnetic, alloy formation, interistitial compounds formation. Interistitial compounds are those compounds in which small atoms like carbon and boron fits into interistices of d-block elements crystal. In interistitial compounds, there is no chemical bond formation takes place so , chemical properties remain almost same but physical properties may change.

7. Which of the property of interistitial compounds has the same behaviour as that of the element -

(A) Malleability (B) Ductility

(C) Electrical conductance (D) Hardness

DB0077

8. Which of the following property gets decreased in interistitial compounds compared to that of the element -

(A) Malleability (B) Metallic lustre

(C) Hardness (D) Density

DB0077

9. Select correct statement -

(A) Highest oxidation state of 3d-series is +8.

(B) Ni, Cu and Zn are not transition element.

(C) Ziglar natta catalyst contain vanadium.

(D) Aq. solution of Cu^{2+}, Fe^{+3} and Cr^{3+} are blue, yellow and green respectively.

DB0077

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MATCH THE COLUMN : (Matrix Match)

10.

Column-I (Metals)	**Column-II**
(A) Zn	(P) Cyanide process involve in the comerical extration
(B) Cu	(Q) Extracted by hydrometallurgical process
(C) Ag	(R) Roasting involve in the comerical extration
(D) Au	(S) Present in Brass

DB0078

11.

Column-I (catalyst)	**Column-II (Used)**
(A) $TiCl_4$	(P) Adams catalyst in reduction
(B) $PdCl_2$	(Q) In preparation of $(CH_3)_2$ $SiCl_2$
(C) Pt/PtO	(R) Used as the ziegler-natta catalyst in polythene production
(D) Cu	(S) Wacker process for converting C_2H_4 to CH_3CHO

DB0079

SELECT CORRECT CODE :

12.

Column-I	**Column-II**
(P) Cr_2O_3	(1) Neutral oxide
(Q) CrO_3	(2) Amphoteric oxide
(R) Fe_3O_4	(3) Mix oxide
(S) N_2O	(4) Acidic oxide

Select correct code for matching -

Code :

	P	Q	R	S
(A)	2	4	3	1
(B)	2	3	4	1
(C)	4	2	3	1
(D)	4	3	1	2

DB0080

13.

Column-I (Metal ion of 3d-series)	**Column-II (Characterstic)**
(P) Ni^{2+}	(1) produce blue aq. solution
(Q) Cr^{2+}	(2) half filled t_{2g} orbitals in octahedral complex
(R) V^{2+}	(3) diamagnetic ion
(S) Ti^{4+}	(4) calculated μ = 2.84 B.M. (spin only)

Select correct code for matching -

Code :

	P	Q	R	S
(A)	4	2	1	3
(B)	3	1	2	4
(C)	4	1	2	3
(D)	1	2	4	3

DB0081

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Sub Ano+Heating Eff+s,d Block-Ron of Salt Ano-(OC)\Eng\04_d-Block\01_Th+Ex.p65

ASSERATION & REASONING :

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

14. **Assertion** : $KMnO_4$ is purple in colour due to charge transfer.

Reason : In MnO_4^-, there is no electron present in d-orbitals of manganese.

DB0082

15. **Assertion** : K_2CrO_4 has yellow colour due to charge transfer.

Reason : CrO_4^{2-} ion is tetrahedral in shape.

DB0083

16. **Assertion** : The highest oxidation state of chromium in its compounds is +6.

Reason : Chromium atom has only six electrons in ns and (n–1) d orbitals.

DB0084

17. **Assertion** : CrO_3 reacts with HCl to form chromyl chloride gas.

Reason : Chromyl chloride (CrO_2Cl_2) has tetrahedral shape.

DB0085

18. **Assertion** : Zinc does not show characteristic properties of transition metals.

Reason : In zinc outermost shell is completely filled.

DB0086

19. **Assertion** : Tungsten has a very high melting point.

Reason : Tungsten is a covalent compound.

DB0087

20. **Assertion** : Equivalent mass of $KMnO_4$ is equal to one-third of its molecular mass when it acts as an oxidising agent in an alkaline medium.

Reason : Oxidation number of Mn is +7 in $KMnO_4$.

DB0088

21. **Assertion** : Cu^+ ion is colourless.

Reason : Four water molecules are coordinated to Cu^+ ion in water.

DB0089

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Soll And-Heating Effs,d-Block-Ron of Soll And-(OC)\Eng\04_d-Block\01_Th+Ex.p65

MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)

The following column 1, 2, 3 represent the different metals of 3d series and their properties
Answer the questions that follow

Column-1 - Hexa aqua complex of dipositive ion of given metal
Column-2 - Magnetic moment (spin only)
Column-3 - Colour

Column - 1 Hexa aqua complex of dipositive ion of given metal	Column - 2 Magnetic moment (spin only)	Column - 3 Colour
(I) Co	(i) $\sqrt{8}$ B.M.	(P) Green
(II) Ni	(ii) $\sqrt{3}$ B.M.	(Q) Pink
(III) V	(iii) $\sqrt{15}$ B.M.	(R) Colourless
(IV) Zn	(iv) Zero	(S) Violet
		(T) Blue

22. Select **CORRECT** combination for metal which have same magnetism as in $[Ti(H_2O)_6]^{4+}$

(A) (II), (iv), (R) (B) (I), (iv), (R) (C) (IV), (iv), (R) (D) (IV), (iv), (T)

DB0090

23. Which is **CORRECT** magnetic moment and colour combination of the product when $[Co(H_2O)_6]^{+2}$ is react with conc. HCl solution

(A) (ii), (Q) (B) (ii), (T) (C) (iii), (Q) (D) (iii), (T)

DB0090

24. Select **CORRECT** combination for hexaaqua complex of metal ion M^{+2} which have half filled t_{2g} configuration (i.e. t_{2g}^3)

(A) (III), (iii), (S) (B) (III), (iii), (P) (C) (III), (ii), (S) (D) (III), (iv), (P)

DB0090

25. $[M(H_2O)_6]^{+2}$ of a metal in **column-1** have combination (i), (P) in **column-2, 3**. Select **CORRECT** combination for hexaammine complex of same metal ion.

(A) (i), (T) (B) (i), (S) (C) (iv), (R) (D) (iv) (T)

DB0090

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\d-Sub Ano-Heating Eff+s,d-Block-Ron of Salt Ano-(OC)\Eng\04_d-Block\01_Th+Ex.p65

EXERCISE # JEE-MAIN

1. Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is incorrect for this reaction : **[AIEEE-2004]**

(1) Evolved I_2 is reduced (2) CuI_2 is formed

(3) $Na_2S_2O_3$ is oxidised (4) Cu_2I_2 is formed

DB0092

2. In context with the transition elements, which of the following statements is incorrect? **[AIEEE-2009]**

(1) In the highest oxidation states of the first five transition elements (Sc to Mn), all the 4s and 3d electrons are used for bonding.

(2) Once the d^5 configuration is exceeded, the tendency to involve all the 3d electrons in bonding decreases.

(3) In addition to the normal oxidation states, the zero oxidation state is also shown by these elements in complexes.

(4) In the highest oxidation states, the transition metal show basic character and form cationic complexes.

DB0093

3. Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is **incorrect** ? **[AIEEE-2012]**

(1) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.

(2) Ferrous oxide is more basic in nature than the ferric oxide.

(3) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.

(4) Ferrous compounds are less volatile than the corresponding ferric compounds.

DB0094

4. Which of the following arrangements does not represent the correct order of the property stated against it ? **[JEE MAIN-2013]**

(1) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour

(2) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size

(3) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution

(4) Sc < Ti < Cr < Mn : number of oxidation states

DB0096

5. Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown - red vapours of: **[JEE MAIN-2013, Online]**

(1) CrO_3 (2) Cr_2O_3 (3) $CrCl_3$ (4) CrO_2Cl_2

DB0097

6. The element with which of the following outer electron configuration may exhibit the largest number of oxidation states in its compounds : **[JEE MAIN-2013, Online]**

(1) $3d^74s^2$ (2) $3d^84s^2$ (3) $3d^54s^2$ (4) $3d^64s^2$

DB0098

7. When a small amount of $KMnO_4$ is added to concentrated H_2SO_4, a green oily compound is obtained which is highly explosive in nature. Compound may be : **[JEE MAIN-2013, Online]**

(1) Mn_2O_3 (2) $MnSO_4$ (3) Mn_2O_7 (4) MnO_2

DB0099

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Salt Anal-Heating Effs.,d-Block-Ron of Salt Anal(OC)\Eng\04_d-Block\01_Th+Ex.p65

8. Which series of reactions correctly represents chemical relations related to iron and its compound?

(1) $Fe \xrightarrow{Cl_2,\ heat} FeCl_3 \xrightarrow{heat,\ air} FeCl_2 \xrightarrow{Zn} Fe$

(2) $Fe \xrightarrow{O_2,\ heat} Fe_3O_4 \xrightarrow{CO,\ 600°C} FeO \xrightarrow{CO,700°C} Fe$

(3) $Fe \xrightarrow{dil\ H_2SO_4} FeSO_4 \xrightarrow{H_2SO_4,\ O_2} Fe_2(SO_4)_3 \xrightarrow{Heat} Fe$

(4) $Fe \xrightarrow{O_2,\ heat} FeO \xrightarrow{dil\ H_2SO_4} FeSO_4 \xrightarrow{Heat} Fe$ **[JEE MAIN-2014]**

DB0100

9. Which of the following is **not** formed when H_2S reacts with acidic $K_2Cr_2O_7$ solution ?

[JEE MAIN-2014, Online]

(1) K_2SO_4 (2) $Cr_2(SO_4)_3$ (3) S (4) $CrSO_4$ **DB0102**

10. Copper becomes green when exposed to moist air for a long period. This is due to :-

[JEE MAIN-2014, Online]

(1) the formation of a layer of cupric oxide on the surface of copper.

(2) the formation of basic copper sulphate layer on the surface of the metal

(3) the formation of a layer of cupric hydroxide on the surface of copper.

(4) the formation of a layer of basic carbonate of copper on the surface of copper.

DB0103

11. Which one of the following exhibits the largest number of oxidation states ?

[JEE MAIN-2014, Online]

(1) Mn(25) (2) V(23) (3) Cr (24) (4) Ti (22)

DB0104

12. Match the catalysts to the correct processes :- **[JEE MAIN-2015]**

	Catalyst		Process
(A)	$TiCl_3$	(i)	Wacker process
(B)	$PdCl_2$	(ii)	Ziegler-Natta polymerization
(C)	$CuCl_2$	(iii)	Contact process
(D)	V_2O_5	(iv)	Deacon's process

(1) A-ii, B-iii, C-iv, D-i (2) A-iii, B-i, C-ii, D-iv

(3) A-iii, B-ii, C-iv, D-i (4) A-ii, B-i, C-iv, D-iii

DB0107

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Ana+Heating Eff+s_d-Block+Ion of Salt Ana+(OC)\Eng\04_d-Block\01_Th+Ex.p65

13. Which of the following statements is false :- **[JEE MAIN-2015, Online]**

(1) $Cr_2O_7^{2-}$ has a Cr – O – Cr bond

(2) CrO_4^{2-} is tetrahedral in shape

(3) $Na_2Cr_2O_7$ is a primary standard in volumetry

(4) $K_2Cr_2O_7$ is less soluble than $Na_2Cr_2O_7$

DB0108

14. The transition metal ions responsible for colour in ruby and emerald are, respectively : **[Jee Main 2016]**

(1) Cr^{3+} and Cr^{3+} (2) Cr^{3+} and Co^{3+} (3) Co^{3+} and Cr^{3+} (4) Co^{3+} and Co^{3+}

DB0109

15. Galvanization is applying a coating of :- **[Jee Main 2016]**

(1) Zn (2) Pb (3) Cr (4) Cu

DB0110

16. Which of the following compounds is metallic and ferromagnetic ? **[Jee Main 2016]**

(1) MnO_2 (2) TiO_2 (3) CrO_2 (4) VO_2

DB0111

17. Which one of the following species is stable in aqueous solution? **[Jee Main 2016]**

(1) MnO_4^{3-} (2) MnO_4^{2-} (3) Cu^+ (4) Cr^{2+}

DB0112

18. What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$?

(1) The copper metal will dissolve and zinc metal will be deposited **[Jee Main 2016]**

(2) No reaction will occur

(3) The copper metal will dissolve with evolution of oxygen gas

(4) The copper metal will dissolve with evolution of hydrogen gas

DB0113

19. Which of the following ions does **not** liberate hydrogen gas on reaction with dilute acids? **[JEE MAIN-2017, Online]**

(1) Ti^{2+} (2) Cr^{2+} (3) Mn^{2+} (4) V^{2+}

DB0114

20. In the following reactions, ZnO is respectively acting as a/an : **[JEE MAIN-2017]**

(a) $ZnO + Na_2O \rightarrow Na_2ZnO_2$

(b) $ZnO + CO_2 \rightarrow ZnCO_3$

(1) base and acid (2) base and base

(3) acid and acid (4) acid and base

DB0115

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal-Heating Effs,d-Block-Ron of Salt Anal-(OC)\Eng\04_d-Block\01_Th+Ex.p65

21. Conside the following reactions : **[JEE MAIN-2020]**

$NaCl + K_2Cr_2O_7 + H_2SO_4(Conc.) \rightarrow (A) + Side\ products$

$(A) + NaOH \rightarrow (B) + Side\ product$

$(B) + H_2SO_4(dilute) + H_2O_2 \rightarrow (C) + Side\ product$

The sum of the total number of atoms in one molecule each of (A), (B) and (C) is

DB0133

22. The incorrect statement(s) among (a) - (c) is (are) :- **[JEE MAIN-2020]**

(a) W(VI) is more stable than Cr(VI).

(b) in the presence of HCl, permanganate titrations provide satisfactory results.

(c) some lanthanoid oxides can be used as phosphors.

(1) (a) and (b) only (2) (a) only

(3) (b) and (c) only (4) (b) only

DB0134

23. Fex_2 and Fey_3 are known when x and y are : **[JEE MAIN-2021]**

(1) x = F, Cl, Br, I and y = F, Cl, Br

(2) x = F, Cl, Br and y = F, Cl, Br, I

(3) x = Cl, Br, I and y = F, Cl, Br, I

(4) x = F, Cl, Br, I and y = F, Cl, Br, I

DB0135

24. Given below are two statements: **[JEE MAIN-2021]**

Statement I : Potassium permanganate on heating at 573 K forms potassium manganate.

Statement II : Both potassium permanganate and potassium manganate are tetrahedral and paramagnetic in nature.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is true but statement II is false

(2) Both statement I and statement II are true

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are false

DB0136

25. Given below are two statements: **[JEE MAIN-2022]**

Statement I : Iron (III) catalyst, acidified $K_2Cr_2O_7$ and neutral $KMnO_4$ have the ability to oxidise I^- to I_2 independently.

Statement II: Manganate ion is paramagnetic in nature and involves $p\pi$ -$p\pi$ bonding.

In the light of the above statements, choose the correct answer from the options.

(1) Both statement I and Statement II are ture

(2) Both statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

26. The number of terminal oxygen atoms present in the product B obtained from the following reaction is _____. **[JEE MAIN-2022]**

$FeCr_2O_4 + Na_2CO_3 + O_2 \rightarrow A + Fe_2O_3 + CO_2$

$A + H^+ \rightarrow B + H_2O + Na^+$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Ana+Heating Eff+s_d-Block+Ion of Salt Ana+(IOC)\Eng\04_d-Block\01_Th+Ex.p65

EXERCISE # JEE-ADVANCED

TRUE/FALSE :

1. Cu^+ disproportionates to Cu^{2+} and elemental copper in solution. **[JEE 1991]**

DB0116

FILL IN THE BLANKS :

2. Silver jewellery items tarnish slowly in the air due to their reaction with ..... . **[JEE 1997]**

DB0117

MULTIPLE CHOICE QUESTIONS WITH ONE CORRECT ANSWER :

3. Which one is solder ? **[JEE 1995]**

(A) Cu and Pb (B) Zn and Cu (C) Pb and Sn (D) Fe and Zn

DB0118

4. Ammonium dichromate is used in some fireworks. The green coloured powder blown in the air is - **[JEE 1997]**

(A) CrO_3 (B) Cr_2O_3 (C) Cr (D) $CrO(O_2)$

DB0119

5. The number of moles of $KMnO_4$ that will be needed to react with one mole of sulphite ion in acidic solution is - **[JEE 1997]**

(A) 2/5 (B) 3/5 (C) 4/5 (D) 1

DB0120

6. Read the following statement and explanation and answer as per the options given below : **[JEE 1998]**

Assertion : Zn^{2+} is diamagnetic.

Reason : Two electrons are lost from 4s orbital to form Zn^{2+}.

(A) If both assertion and reason are correct, and reason is the correct explanation of the assertion.

(B) If both assertion and reason are correct, but reason is not the correct explanation of the assertion.

(C) If assertion is correct but reason is incorrect.

(D) If assertion is incorrect but reason is correct.

DB0121

7. In the dichromatic anion, **[JEE 1999]**

(A) 4 Cr – O bonds are equivalent (B) 6 Cr – O bonds are equivalent

(C) all Cr – O bonds are equivalent (D) all Cr – O bonds are non-equivalent

DB0122

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal-Heating Effs,d-Block-Ron of Salt Anal(OC)\Eng\04_d-Block\01_TH+Ex.p65

8. Anhydrous ferric chloride is prepared by: **[JEE 2002]**

(A) heating hydrated ferric chloride at a high temperature in a stream of air

(B) heating metallic iron in a stream of dry chlorine gas

(C) reaction of ferric oxide with HCl

(D) reaction of metallic iron with HCl

DB0123

9. When MnO_2 is fused with KOH, a coloured compound is formed, the product and its colour is:

(A) K_2MnO_4, green (B) $KMnO_4$, purple **[JEE 2003]**

(C) Mn_2O_3, brown (D) Mn_3O_4, black

DB0124

10. The product of oxidation of I^- with MnO_4^- in alkaline medium is - **[JEE 2004]**

(A) IO_3^- (B) I_2 (C) IO^- (D) IO_4^-

DB0125

11. $(NH_4)_2Cr_2O_7$ on heating liberates a gas. The same gas will be obtained by - **[JEE 2004]**

(A) heating NH_4NO_2 (B) heating NH_4NO_3

(C) treating H_2O_2 with $NaNO_2$ (D) treating Mg_3N_2 with H_2O

DB0126

12. Which of the following combination will produce H_2 gas ? **[JEE Advance 2017]**

(A) Zn metal and NaOH(aq) (B) Au metal and NaCN(aq) in the presence of air

(C) Cu metal and conc. HNO_3 (D) Fe metal and conc. HNO_3

DB0127

MULTIPLE CHOICE QUESTIONS WITH ONE OR MORE THAN ONE CORRECT ANSWER :

13. Which of the following alloys contains (s) Cu and Zn ? **[JEE 1993]**

(A) Bronze (B) Brass (C) Gun metal (D) Type metal

DB0128

14. Addition of high proportions of magnanese makes steel useful in making rails of railroads, becuse manganese. **[JEE 1998]**

(A) gives hardness to steel

(B) helps the formation of oxides of iron

(C) can remove oxygen and sulphur

(D) can show highest oxidation state of +7.

DB0129

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15. The correct statement(s) about Cr^{2+} and Mn^{3+} is (are) **[JEE Advance 2015]**

[Atomic numbers of Cr = 24 and Mn = 25]

(A) Cr^{2+} is a reducing agent

(B) Mn^{3+} is an oxidizing agent

(C) Both Cr^{2+} and Mn^{3+} exhibit d^4 electronic configuration

(D) When Cr^{2+} is used as a reducing agent, the chromium ion attains d^5 electronic configuration

DB0130

16. Fusion of MnO_2 with KOH in presence of O_2 produces a salt **W**. Alkaline solution of **W** upon eletrolytic oxidation yields another salt **X**. The manganese containing ions present in **W** and **X**, respectively, are **Y** and **Z**. Correct statement(s) is (are) **[JEE Advance 2019]**

(A) **Y** is diamagnetic in nature while **Z** is paramagnetic

(B) Both **Y** and **Z** are coloured and have tetrahedral shape

(C) In both **Y** and **Z** , π-bonding occurs between p-orbitals of oxygen and d-orbitals of manganese.

(D) In aqueous acidic solution, **Y** undergoes disproportionation reaction to give **Z** and MnO_2.

DB0131

17. Consider the following reactions (unbalanced) **[JEE Advance 2019]**

$Zn + \text{hot conc. } H_2SO_4 \rightarrow G + R + X$

$Zn + \text{conc. } NaOH \rightarrow T + Q$

$G + H_2S + NH_4OH \rightarrow Z \text{ (a precipitate)} + X + Y$

Choose the correct option(s).

(A) The oxidation state of Zn in T is +1

(B) Bond order of Q is 1 in its ground state

(C) Z is dirty white in colour

(D) R is a V-shaped molecule

DB0132

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Soll Anal-Heating Effs,d-Block-Ron of Soll Anal-(OC)\Eng\04_d-Block\01_Th+Ex.p65

ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	A	B	B	B	D	A	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	A	C	B	BC	B	A	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	B	A	B	C	B	B	D	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	AC	D	D	C	B	C	A	C	D
Que.	41	42	43	44	45	46	47	48		
Ans.	A	A	B	A	A	B	B	D		

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABC	BCD	AB	ACD	AB	AB	ABD	BC	ABC	ABC
Que.	11	12	13	14	15	16	17	18		
Ans.	AB	ABCD	AB	AB	BC	ABC	ABC	ABC		

EXERCISE # S-1

1. **(6),** In this reaction CrO_2Cl_2 is formed in which Cr is +6.

2. **(2),** V^{+4}, Cu^{+2}

3. **(11),** +7, +4

4. **(3),** Mn^{+3}, Fe^{+3}, Co^{+3}

5. CrO_2Cl_2 has d^3S hybridisation and all 3d-orbitals are non-axial which are d_{xy} ; d_{yz} and d_{xz}.

6. **(003)**

7. **(6),** Mohr's salt

$FeSO_4 .(NH_4)_2SO_4.6H_2O$

$6\ Fe^{+2} + Cr_2O_7^{2-} + 14\ H^+ \rightarrow 2Cr^{+3} + 7H_2O + 6Fe^{+3}$

8. **(3)**

(I) $BaCl_2 + Na_2CrO_4 \xrightarrow{CH_3COOH} BaCrO_4 \downarrow + 2NaCl$

(II) $K_2Cr_2O_7 + NaOH \longrightarrow K_2CrO_4$

$CrO_4^{2-} \underset{OH^-}{\overset{H^+}{\rightleftharpoons}} Cr_2O_7^{2-}$

(III) $Cr_2O_7^{2-} + NO_3^- \xrightarrow{H^+}$ no reaction.

(IV) $Cr_2O_7^{2-} + C_2H_5OH \xrightarrow{H^+} CH_3 - COOH + Cr^{+3}$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Ana+Heating Eff+s,d-Block+Ion of Salt Ana+(OC)\Eng\04_d-Block\01_Th+Ex.p65

EXERCISE # S-2

Que.	1	2	3	4	5	6	7	8	9
Ans.	C	D	A	B	B	D	C	A	D
Que.	10				11				12
Ans.	A→RS ; B→RS ; C→PQ ; D→PQ				A→R ; B→S ; C→P ; D→Q				A
Que.	13	14	15	16	17	18	19	20	21
Ans.	C	B	B	A	B	C	C	B	C
Que.	22	23	24	25					
Ans.	C	D	A	A					

EXERCISE # JEE MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	1	1	4	3	3	2	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	3	1	1	3	2	2	3	4
Que.	21	22	23	24						
Ans.	18.00	4	1	1						

25. Ans. (2)

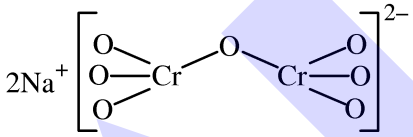
Sol. Neutral $KMnO_4$ oxidises I^- to IO_3^-

Manganate ion has $d\pi$-$p\pi$ bonding.

26. Ans. (6)

Sol. $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$

$$2Na_2CrO_4 + 2H^+ \rightarrow \underbrace{Na_2Cr_2O_7}_{B} + 2Na^+ + H_2O$$



EXERCISE # JEE ADVANCE

TRUE/FALSE :

1. True

FILL IN THE BLANKS :

2. H_2S

Que.	3	4	5	6	7	8	9	10	11	12
Ans.	C	B	A	B	B	B	A	A	A	A
Que.	13	14	15							
Ans.	BC	A	ABC							

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal-Heating Effs,d-Block-Ron of Salt Anal(OC)\Eng\04_d-Block\01_Th+Ex.p65

16. Ans.(B,C,D)

$$MnO_2 + 2KOH + \frac{1}{2}O_2 \xrightarrow{\Delta} \underset{(W)}{K_2MnO_4} + H_2O$$

$$\left[(W) = K_2MnO_{4(aq)} \rightleftharpoons 2K^{\oplus}_{(aq)} + \underset{(Y)}{MnO^{2-}_{4(aq)}}\right]$$

$$K_2MnO_4 + H_2O \xrightarrow[\text{Redox}]{\text{Electolytic}} H_2 + KOH + \underset{(X)}{KMnO_4}$$

[anion of X = $\underset{(Z)}{MnO_4^-}$]

$$\left[\because \underset{(Y)}{MnO_4^{2-}} \xrightarrow[\text{Oxidation}]{\text{Electrolytic}} \underset{(Z)}{MnO_4^-} + e^-\right]$$

$\because$ In acidic solution; Y undergoes disproportionation reaction

$$\left[3\underset{(Z)}{MnO^{2-}_{4(aq)}} + 4H^{\oplus} \longrightarrow 2MnO_4^- + MnO_2 + 2H_2O\right]$$

17. Ans.(B,C,D)

Sol. $Zn + 2H_2SO_4$ (Hot and conc.) $\rightarrow \underset{(G)}{ZnSO_4} + \underset{(R)}{SO_2}\uparrow + \underset{(X)}{2H_2O}$

$Zn + 2NaOH$ (conc.) $\rightarrow \underset{(T)}{Na_2ZnO_2} + \underset{(Q)}{H_2}\uparrow$

$ZnSO_4 + H_2S + 2NH_4OH \rightarrow \underset{(Z)}{ZnS}\downarrow + \underset{(X)}{2H_2O} + \underset{(Y)}{(NH_4)_2SO_4}$

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